

Seyed Mohammadreza Alavi

School of Electrical and Computer Engineering - University of Tehran

 [Website](#)  [Github](#)  [LinkedIn](#)  [Email](#)  [Google Scholar](#)

EDUCATION

B.Sc. in Computer Engineering

Sep 2021 - Present

University of Tehran

GPA: 18.7/20

High School Diploma in Mathematics

Sep 2018 - May 2021

Rouzbeh High School

GPA: 19.56/20

RESEARCH INTERESTS

- Artificial Intelligence (Machine Learning, Deep Learning, Neural Networks, Natural Language Processing, Generative Models, Algorithmic Recourse)
- Data Science
- Cryptography, Blockchain, and Cryptocurrencies

CERTIFICATIONS

Deep Learning Specialization ([Certificate](#))

Coursera - DeepLearning.AI

Machine Learning Specialization ([Certificate](#))

Coursera - DeepLearning.AI - Stanford University

Machine Learning with Python ([Certificate](#))

Maktabkhone

Blockchain Specialization ([Certificate](#))

Coursera - University at Buffalo

Introduction to Blockchain & Bitcoin ([Certificate](#))

Maktabkhoneh

Advanced Python Programming ([Certificate](#))

Maktabkhone

Network Principles ([Certificate](#))

Maktabkhone

ICDL ([Certificate](#))

ICDL Institute

ACADEMIC EXPERIENCE

Remote Research Assistant

University of Waterloo

Prof. Amir-Hossein Karimi

Research Assistant

Tehran Institute for Advanced Studies (TEIAS) - Khatam University

Prof. Behnam Bahrak

Supervisor

Artificial Intelligence Course

Prof. Yadollah Yaghoobzadeh and Dr. Hakimeh Fadaei

Teaching Assistant

Artificial Intelligence Course

Prof. Yadollah Yaghoobzadeh and Dr. Hakimeh Fadaei

Teaching Assistant

Data Science Course

Prof. Yadollah Yaghoobzadeh and Prof. Behnam Bahrak

Teaching Assistant

Engineering Probability and Statistics

Prof. Mostafa Tavassolipour and Prof. Abdole-hossein Vahabie

Teaching Assistant

Engineering Probability and Statistics

Prof. Behnam Bahrak and Prof. Abdole-hossein Vahabie

Teaching Assistant

Introduction to Computing Systems and Programming Course

Prof. Mohammadreza Hashemi and Prof. Hadi Moradi

Instructor

Python and Algorithm Course

Rouzbeh High School

SELECTED ACADEMIC PROJECTS

Star Wars

This project is a simulation of graphical star wars game that a ship tries to eliminate enemies and saves the hostage.

U-Taxi

This project is a simulation of an online taxi service that can be run on local host.

Snake Game

This project is a simulation of graphical snake game that a snake tries to eat mice.

Artificial Intelligence Projects

These projects are AI course projects include: Genetic Algorithm, Reinforcement Learning, Hidden Markov Model, Regression and Classification, Convolutional Neural Networks, and Clustering.

RISC-V Processor

These projects include RISC-V single cycle, RISC-V multicycle, and RISC-V pipeline.

Xv6 Operating System Projects

These projects include some modifications to the Xv6 operating system and we have added some features to this operating system.

Tiny Me

This project is a matching engine for a stock market that we have completed.

Function Craft

Function Craft is a functional programming language that we have designed and implemented in four phases.

Voice Call

This project is a distributed live voice call system using WebRTC and Qt framework. It's a peer-to-peer audio communication platform that captures and transmits real-time voice data between two peers.

Network Simulator

This project is a simulation of network functionality with different routing algorithms like RIP and OSPF in various network topologies

PUBLICATIONS

Generating Bearing Vibration Signals at User-Specified Fault Probabilities Using PR-GAN and Counterfactual Methods

Jul. 2026

Seyed Mohammadreza Alavi, Ardeshir Shojaeinasab, Reza Jalayer, Masoud Jalayer, and Behnam Bahrak
arXiv preprint, arXiv:2607.19455 [cs.LG]

This study develops PR-GAN and counterfactual methods to generate bearing vibration signals with user-defined fault probabilities. The methods are evaluated on the CWRU and Paderborn datasets based on probability accuracy and signal similarity.

SKILLS

Python, C/C++, HTML/CSS, Verilog/SystemVerilog, Algorithm, Data Structure, Artificial Intelligence, Machine Learning, Deep Learning, Neural Networks, Matlab, IC DL, Linux, Java, SQL, Database, Compiler, Cryptography, Blockchain, Cryptocurrencies, Algorithmic Recourse, LaTeX, Git, Data Science, Networking

SELECTED UNIVERSITY COURSES

Advanced Programming	20/20
Engineering Probability and Statistics	20/20
Data Structures and Algorithms	19/20
Algorithm Design	19/20
Artificial Intelligence	19.5/20
Data Science	18.5/20
Cryptocurrencies (<i>Voluntary Graduate-Level Course</i>)	17.1/20
Additional Courses	
Technical English	20/20
Research and Technical Presentation	20/20

LANGUAGES

Persian (Native)
English